

TA Recruitment System — User Manual

Group 48 · EBU6304 Software Engineering Group Project · May 2026

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1. Getting Started

Prerequisites

- Java 17+

- Apache Maven 3.9+
- Apache Tomcat 9.x (or use embedded Jetty for local development)

Quick Start (Jetty)

```
cd EBU6304-Group48
mvn jetty:run
```

Open **<http://localhost:8080/ta-recruitment/>** in your browser. If port 8080 is in use:

```
mvn jetty:run -Djetty.port=9080
```

Production Deploy (Tomcat)

```
mvn package
# Copy target/ta-recruitment.war to Tomcat webapps/
# Start Tomcat
# Open http://localhost:8080/ta-recruitment/
```

Data Storage

All data is stored as JSON files under `${user.home}/ebu6304-group48-data/`. On first startup the demo dataset is automatically seeded from the bundled WAR resources.

2. Demo Accounts

Username	Role	Password
ta_demo	TA	demo123
ta_li	TA	demo123
mo_demo	MO	demo123
admin_demo	ADMIN	demo123

You can also register new accounts via the **Register** page.

3. TA Guide

3.1 TA Dashboard

After logging in as a TA, you land on the dashboard. It shows:

- **Stats bar:** number of open jobs, your applications, in-review count, pending invitations
- **Invitation alert:** a prominent banner if you have pending MO invitations
- **Profile completeness card:** shown if your profile is under 100%
- **Latest applications:** recent 5 applications with status badges and match scores
- **Best Matches for You:** open jobs ranked by match score, with matched/missing skills

The screenshot displays the TA Recruitment dashboard for a user named 'ta_demo'. The interface features a dark blue header with the title 'TA Recruitment' and a 'Logout' button. Below the header is a navigation bar with tabs for 'Dashboard', 'Profile', 'CV', 'Jobs', 'Status', and 'Invitations'. The main content area is divided into several sections:

- Welcome back, ta_demo:** A greeting section showing statistics: 3 Open, 2 Applied, 1 In Review, and 1 Invitation.
- Invitation Alert:** A light blue banner stating 'You have 1 pending invitation from MOs.' with a 'View Invitations →' button.
- Your Profile:** A card showing '83% complete' with 'Edit Profile' and 'Upload CV' buttons.
- Latest Applications:** A list of recent applications with status badges and match scores:
 - Final Exam Invigilation (Hall A): UNDER_REVIEW, 63% match, 2026-04-11 14:30
 - Software Engineering Lab TA: SUBMITTED, 89% match, 2026-04-11 14:00
- Best Matches for You:** A list of recommended jobs with match scores and skill tags:
 - Software Engineering Lab TA: 89% match, Apply →. Skills: Java, Teaching, Algorithms.
 - Final Exam Invigilation (Hall A): 63% match, Apply →. Skills: Teaching, Proctoring.
 - Database Systems Tutorial: 47% match, Apply →. Skills: Teaching, Java, Database.

At the bottom of the dashboard, there are buttons for 'My Status', 'Edit Profile', and 'Upload CV'.

3.2 TA Profile

Navigate to **Profile** in the top nav. Fill in:

- 1. **Full Name** (required)
- 2. **Email Address** (required)
- 3. **Major/Program** (required)
- 4. **Skills** — type comma-separated values (e.g. "Java, Python, Teaching").
Suggested skills from open jobs appear as clickable chips.
- 5. **Availability** — check all time slots you are available (20 options matching MO schedule choices).
- 6. **Additional Notes** (optional)

Click **Save profile**. A success message appears at the top.

TA Recruitment

Logout

Dashboard

Profile

CV

Jobs

Status

Invitations

← Back to Dashboard

TA Profile Management

Personal Information

Full Name *

Alex Chen

Email Address *

alex.chen.demo@example.edu

Major/Program *

Software Engineering

Skills (comma-separated)

Java, Python, Web Development, Teaching, Algorithms

Suggested skills from open jobs:

DatabaseTeachingJavaProctoringAlgorithms

Availability (select available time slots)

☐ Mon 09:00-12:00

☒ Mon 14:00-16:00

☐ Mon 18:00-20:00

☐ Tue 09:00-12:00

☐ Tue 14:00-16:00

☐ Tue 18:00-20:00

☐ Wed 09:00-12:00

☐ Wed 14:00-16:00

☒ Wed 18:00-20:00

☐ Thu 09:00-12:00

☐ Thu 14:00-16:00

☐ Thu 18:00-20:00

☐ Fri 09:00-12:00

☒ Fri 10:00-12:00

☐ Fri 14:00-16:00

☐ Fri 18:00-20:00

☐ Sat 09:00-12:00

☐ Sat 14:00-16:00

☐ Sun 09:00-12:00

☐ Sun 14:00-16:00

Additional Notes

Demo TA profile — prefers lab and tutorial sessions.

Save profile

Last updated: 2026-05-23 14:37

3.3 TA CV Management

Navigate to **CV** in the top nav.

- **Upload CV:** Click "Choose File", select a `.txt` file, then click "Upload CV". The system extracts skills via AI analysis.
- **Set as Active:** If you have multiple CVs, click "Set as Active" on the one you want to use for applications.
- **Delete CV:** Click "Delete" to remove an old CV.

The screenshot shows the 'TA Recruitment' dashboard with the 'CV' tab selected in the top navigation. The main heading is 'CV Management'. A yellow box titled 'File requirements' lists: Accepted format: TXT (plain text) only; Maximum file size: 5MB; File naming: CV_UserID_Timestamp.Extension (auto-generated); and You can upload multiple CVs, but only one can be "active" for applications. Below this is the 'Upload New CV' section with a 'Select CV File *' dropdown showing 'Choose File' and 'No file chosen', a note 'Plain text files (.txt) only. Max 5MB. AI will analyze your CV and auto-fill skills.', and an 'Upload CV' button. The 'Your CVs' section shows 'No active CV selected. Please upload a CV to apply for jobs.' and a table of 'All CV Files' with one entry: 'CV_U-DEMO-TA_20260412090000.pdf' with 'Archive', 'Delete', and 'Set as Active' buttons. A 'CV Tips' section at the bottom lists advice on keeping the CV updated, highlighting teaching experience, including programming skills, matching contact info, and having different versions for course types.

TA Recruitment Logout

Dashboard Profile **CV** Jobs Status Invitations

[← Back to Dashboard](#)

CV Management

File requirements

- Accepted format: TXT (plain text) only
- Maximum file size: 5MB
- File naming: CV_UserID_Timestamp.Extension (auto-generated)
- You can upload multiple CVs, but only one can be "active" for applications

Upload New CV

Select CV File *

Choose File No file chosen

Plain text files (.txt) only. Max 5MB. AI will analyze your CV and auto-fill skills.

Upload CV

Your CVs

No active CV selected. Please upload a CV to apply for jobs.

All CV Files

CV_U-DEMO-TA_20260412090000.pdf	Archive	Delete	Set as Active
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CV Tips

- Keep your CV updated with recent experience and skills
- Highlight relevant teaching or tutoring experience
- Include specific programming languages and tools you're proficient in
- Make sure contact information matches your profile
- Consider having different CV versions for different course types (programming vs. theory)

3.4 Browse & Apply for Jobs

Navigate to **Jobs** in the top nav.

- **Filters:** Use the dropdowns and text input to filter by Job Type, Semester, or Required Skill.
- **Job table:** Each row shows title, type, schedule, required skills (matched ✓ / missing ✕), match score percentage, status, and an **Apply** button.
- **AI Analysis:** Click "🌟 AI Analysis" or "🔍 Skill Gap" for a detailed match explanation.
- **Apply:** Click the green "Apply" button on an OPEN job. You must have a complete profile and CV first.

TA Recruitment

Logout

Dashboard
Profile
CV
Jobs
Status
Invitations

Available TA Jobs

5
Skills Listed

3
Time Slots

2
Applications

CV missing. Please upload your CV before applying for jobs. Most applications require a current CV.

Job Search Filters

Job Type

Semester

Required Skill

All Types

All Semesters

e.g., Java, Teaching

Apply Filters

Clear Filters

Available Positions (3)

Job Title	Type	Schedule	Skills Required	Match	Status	Action
Software Engineering Lab TA 2026 Spring • Capacity: 3	MODULE	WED_18_20	Java ✓ Teaching ✓ Algorithms ✓	89% S:44/50 T:25/25 M:15/15 P:5/10 AI Analysis Skill Gap	APPLIED	Applied
Final Exam Invigilation (Hall A) 2026 Spring • Capacity: 4	INVIGILATION	FRI_09_12	Proctoring ✗ Teaching ✓	63% S:28/50 T:15/25 M:15/15 P:5/10 AI Analysis Skill Gap	APPLIED	Applied
Database Systems Tutorial 2026 Spring • Capacity: 2	MODULE	TUE_18_20	Database ✗ Teaching ✓ Java ✓	47% S:32/50 T:0/25 M:10/15 P:5/10 AI Analysis Skill Gap	OPEN	Apply

Showing 3 jobs
Last updated: 2026-04-10 10:00

Application Tips

- Match your skills:** Highlight skills that match the job requirements in your profile
- Check availability:** Ensure your availability matches the job schedule
- Update CV:** Make sure your CV is current and highlights relevant experience
- Apply early:** Popular positions fill up quickly
- Track applications:** Check your application status regularly

View Application Status

3.5 Application Status

Navigate to **Status** in the top nav.

Shows all your submitted applications with: - Job title, application ID, applied date -
Current status: SUBMITTED → UNDER REVIEW → SELECTED / REJECTED - Match

scores and skill breakdowns

The screenshot shows a web application titled "TA Recruitment" with a dark blue header. In the top right corner of the header is a "Logout" button. Below the header is a navigation bar with links: "Dashboard", "Profile", "CV", "Jobs", "Status" (which is highlighted with a dark blue background), and "Invitations".

Below the navigation bar, the main section is titled "My application status" in bold. Underneath this title is a link that says "Refresh for the latest updates.".

The core of the page is a table with the following columns: APPLICATION ID, JOB, MATCH, MISSING SKILLS, STATUS, FEEDBACK, LAST UPDATED, and ACTION.

APPLICATION ID	JOB	MATCH	MISSING SKILLS	STATUS	FEEDBACK	LAST UPDATED	ACTION
A-DEMO-002	Final Exam Invigilation (Hall A)	50% Skills: 28/50 (Teaching matched; Proctoring missing) Schedule: 15/25 (not matched) Major: 15/15 Profile: 5/10	Proctoring	UNDER_REVIEW	Willing to complete proctoring training if selected.	2026-04-12 08:00	
A-DEMO-001	Software Engineering Lab TA	100% Skills: 44/50 (Java, Teaching, Algorithms matched) Schedule: 25/25 (matched) Major: 15/15 Profile: 5/10	None — all skills matched	SUBMITTED	Interested in SE lab; available WED evening.	2026-04-11 14:00	<button>Withdraw</button>

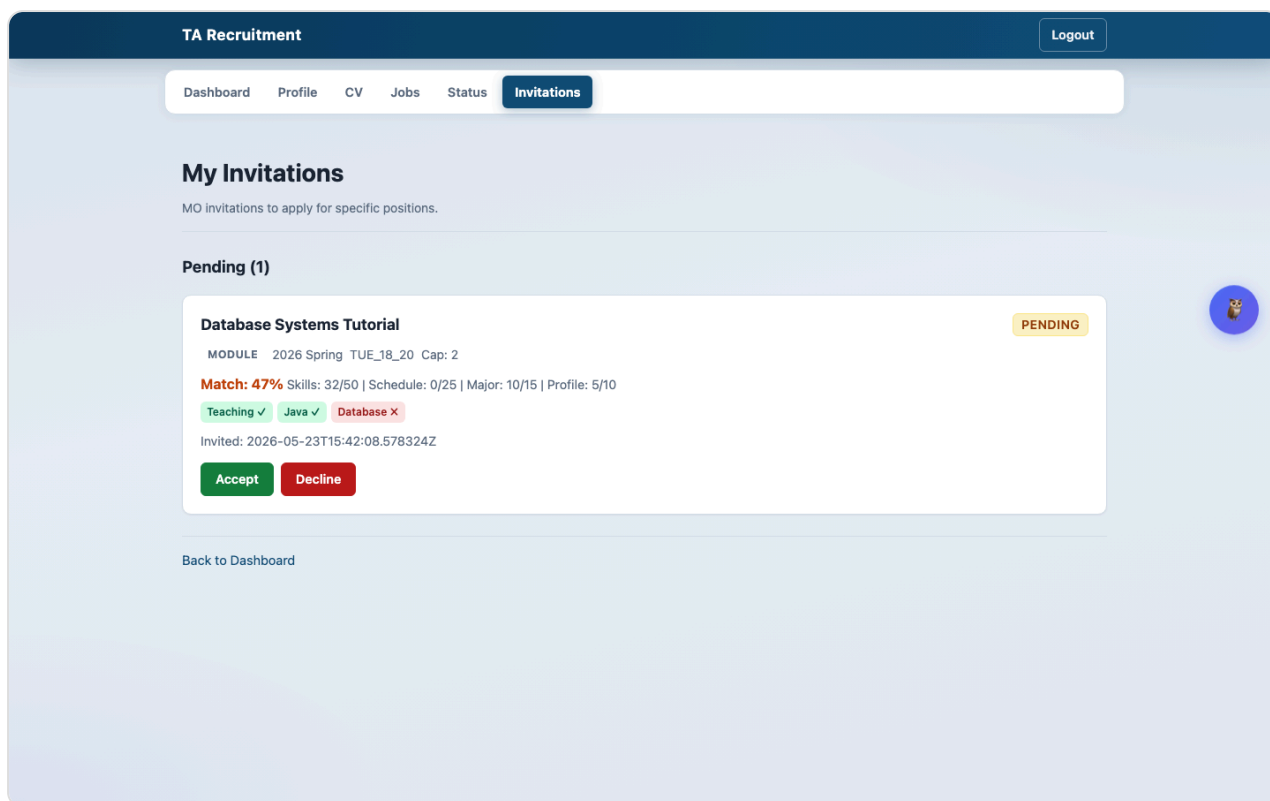
On the right side of the page, there is a circular profile picture placeholder containing a small owl icon.

3.6 Invitations

Navigate to **Invitations** in the top nav.

Shows: - **Pending invitations**: MO-sent invitations with job details, match scores, and Accept/Decline buttons - **History**: Previously accepted or declined invitations

Click **Accept** to automatically create an application. Click **Decline** to reject.



4. MO Guide

4.1 MO Dashboard & Post a Job

After logging in as an MO, you land on the dashboard showing: - **Stats**: total posts, open jobs, pending applications - **Your Positions**: card list of all jobs with applicant counts and status - **Post New Job** button (opens a modal)

Post a New Job

Click "**+ Post New Job**" to open the modal. Fill in:

1. **Title** (required) — e.g. "CS101 Tutorial Assistant"
2. **Type** (required) — MODULE or INVIGILATION
3. **Semester** (required) — e.g. 2026 Spring
4. **Schedule** (required) — 20 time slot options
5. **Capacity** (required) — number of TAs needed
6. **Required Skills** — comma-separated, e.g. "Java, Teaching"
7. **Application Deadline** — optional datetime

Click **"Create job"** to post.

The screenshot displays the 'TA Recruitment' dashboard. At the top, there's a dark blue header with 'TA Recruitment' and a 'Logout' button. Below the header is a navigation bar with 'Dashboard' (active), 'Review applications', 'Invite TAs', and 'Post a new job'. The main section is titled 'My Positions' with a greeting 'Hello, mo_demo'. It features three summary cards: '4 POSTS', '3 OPEN', and '2 PENDING'. Below these are buttons for '+ Post New Job' and 'Review Applications →'. The 'Your Positions' section lists four jobs:

- J-DEMO-003: Database Systems Tutorial** (OPEN)
 - Module: 2026 Spring, Cap: 2
 - 1 applicant
 - Buttons: Review Applicants →, Invite TAs, Close Job
- J-DEMO-002: Final Exam Invigilation (Hall A)** (OPEN)
 - Invigilation: 2026 Spring, Cap: 4
 - 1 applicant
 - Buttons: Review Applicants →, Invite TAs, Close Job
- J-DEMO-001: Software Engineering Lab TA** (OPEN)
 - Module: 2026 Spring, Cap: 3
 - 2 applicants (1 submitted)
 - Buttons: Review Applicants →, Invite TAs, Close Job
- J-DEMO-004: Legacy Module (Closed)** (CLOSED)
 - Module: 2025 Fall, Cap: 1
 - No applicants yet
 - Button: Review Applicants →

4.2 Review Applications

Navigate to **Review Applications** in the top nav.

- **Filter by job:** Select a job from the dropdown
- **Capacity bar:** Shows capacity, selected count, and remaining slots
- **Application table:** Each row shows applicant details (name, major, email, AI CV summary), match breakdown (skills/schedule/major/profile scores), missing skills, and action buttons
- **Batch operations:** Check multiple applicants → sticky bar appears at the bottom → "Mark as Under Review" or "Reject Selected"

Status progression: **SUBMITTED → UNDER REVIEW → SELECTED / REJECTED**

TA Recruitment Logout

[Dashboard](#) **[Review applications](#)** [Invite TAs](#) [Post a new job](#)

MO selection

Filter applications and update status.

Choose job
J-DEMO-003 - Database Systems Tutorial Filter

2
Capacity

1
Selected

1
Remaining

☐	APPLICATION ID	APPLICANT	CV	MATCH BREAKDOWN	MISSING SKILLS	STATUS	ACTION
☐	A-DEMO-004	Li Wei Computer Science li.wei.demo@example.edu	Download CV	Total: 78/100 Skills: 28/50 Schedule: 25/25 (matched) Major: 15/15 Profile: 10/10	Database ✓ Java ✓ Teaching ✗ Skills: 28/50 (Database, Java matched; Teaching missing) Schedule: 25/25 (matched) Major: 15/15 Profile: 10/10	SELECTED	Finalised

0 selected [Mark as Under Review](#) [Reject Selected](#)

Status flow: SUBMITTED → UNDER REVIEW → SELECTED / REJECTED

[Back to MO dashboard](#)

4.3 Invite TAs

Navigate to **Invite TAs** in the top nav.

- Select a job, then browse all TAs ranked by match score
- Click **"Send Invitation"** on promising candidates who haven't yet applied
- TAs receive the invitation on their dashboard and can Accept/Decline

TA Recruitment

Logout

Dashboard

Review applications

Invite TAs

Post a new job

Invite TAs to Job

Browse all TAs ranked by match score and send invitations.

Choose job

J-DEMO-003 - Database Systems Tutorial (OPEN)

Filter

Database Systems Tutorial

Job

3

TAs Available

2

Capacity

TA	PROFILE	MATCH	SKILLS	STATUS	ACTION
ta_li U-DEMO-TA2	Li Wei Computer Science li.wei.demo@example.edu	78/100 S:28/50 T:25/25 M:15/15 P:10/10	Database ✓ Java ✓ Teaching ✗	SELECTED	—
ta_demo U-DEMO-TA	Alex Chen Software Engineering alex.chen.demo@example.edu	47/100 S:32/50 T:0/25 M:10/15 P:5/10	Teaching ✓ Java ✓ Database ✗	INVITED	—
111 U-E5F149CB1F17	No profile	N/A	—	—	No profile

[Back to MO dashboard](#)

5. Admin Guide

5.1 Admin Console (Workload)

After logging in as ADMIN, you land on the workload dashboard.

Left column: - **4 metric cards:** Total Users, Jobs, Applications, Alerts - **Alerts & Warnings:** Over-capacity jobs, open jobs with no selections, rule-based conflict hints - **Per-job Workload Table:** Job ID, status, capacity, application counts by stage, risk

Right column: - **AI Workload Insights:** Click "🌟 Analyze with AI" for AI-generated workload recommendations

Toggle the workload table visibility with the **"Hide Details"** button.

TA Recruitment

Logout

Workload

Users

Applications

Admin Console

Hello, admin_demo

Manage Users

Applications

Hide Details ↑

6

Users

4

Jobs

4

Applications

1

Alerts

AI Workload Insights

Analyze with AI

Alerts & Warnings

Open Jobs Without Selection

2 jobs have no selected TA yet

Per-job Workload

JOB	STATUS	CAPACITY	APPS	SUB	REVIEW	SEL	REJ	REM	RISK
J-DEMO-001 Software Engineering Lab TA	OPEN	3	2	1	0	0	1	3	—
J-DEMO-002 Final Exam Invigilation (Hall A)	OPEN	4	1	0	1	0	0	4	—
J-DEMO-003 Database Systems Tutorial	OPEN	2	1	0	0	1	0	1	—
J-DEMO-004 Legacy Module (Closed)	CLOSED	1	0	0	0	0	0	1	—

5.2 User Management

Navigate to **Users** in the top nav.

- **Role stats:** Total / TA / MO / Banned counts
- **User table:** User ID, username, role (color-coded badge), banned status, ban reason, appeal info
- **Ban/Unban:** Fill in ban reason and click Ban. Click Unban to reverse.
- **Reset Password:** Enter new password and click "Reset PW"

TA Recruitment

Logout

Workload

Users

Applications

User Management

Ban / unban accounts and reset passwords. Password hashes are never shown.

6

Total Users

3

TAs

1

MOs

0

Banned

USER ID	USERNAME	ROLE	BANNED	BAN REASON	APPEAL	ACTIONS
U-E5F149CB1F17	111	TA	NO	—	—	Ban reason... Ban New password Reset PW
U-DEMO-ADMIN	admin_demo	ADMIN	NO	—	—	New password Reset PW
U-D7C9AE0C2233	cqnoxq	ADMIN	NO	—	—	Ban reason... Ban New password Reset PW
U-DEMO-MO	mo_demo	MO	NO	—	—	Ban reason... Ban New password Reset PW
U-DEMO-TA	ta_demo	TA	NO	—	—	Ban reason... Ban New password Reset PW
U-DEMO-TA2	ta_li	TA	NO	—	—	Ban reason... Ban New password Reset PW

5.3 Application Management

Navigate to **Applications** in the top nav.

- **Filter by status:** Dropdown to filter by SUBMITTED / UNDER REVIEW / SELECTED / REJECTED
- **Result count:** "Showing X applications with status Y"
- **Application table:** App ID, job, applicant, match score (color-coded), status badge, admin revoke flag

- **Revoke:** Click "Revoke" on SUBMITTED or UNDER REVIEW applications (SELECTED/REJECTED are finalised)

TA Recruitment

Logout

Workload

Users

Applications

Applications

Review and revoke applications. Finalised applications (SELECTED / REJECTED) cannot be revoked here.

Filter by Status

All

Filter

Showing 4 applications

APPLICATION	JOB	APPLICANT	MATCH	STATUS	ADMIN REVOKE	ACTION
A-DEMO-004	J-DEMO-003 Database Systems Tutorial	U-DEMO-TA2	78%	SELECTED	—	
A-DEMO-003	J-DEMO-001 Software Engineering Lab TA	U-DEMO-TA2	37%	REJECTED	—	
A-DEMO-002	J-DEMO-002 Final Exam Invigilation (Hall A)	U-DEMO-TA	63%	UNDER REVIEW	—	Revoke
A-DEMO-001	J-DEMO-001 Software Engineering Lab TA	U-DEMO-TA	89%	SUBMITTED	—	Revoke

6. Auth Pages

Login

Navigate to `/ta-recruitment/login` .

Enter username and password. Successful login redirects to your role-appropriate landing page.

TA Recruitment

Sign in

Use your course demo account or register a new user.

Username

Password

Sign in

[Create an account](#) [Back to home](#)

Register

Navigate to `/ta-recruitment/register` .

Fill in username, password (min 8 chars), confirm password, and select role (TA or MO). Admin accounts cannot be self-registered.

TA Recruitment

Register

Choose a role for the coursework demo. Passwords are stored as a simple hash only.

Username

Password

Confirm password

Role

TA (applicant) ▾

Create account

[Already have an account](#)

7. Error Pages

404 — Not Found

When navigating to a non-existent page.

404 - Page Not Found

The page you are looking for does not exist or has been moved.

[Back to Home](#)

500 — Internal Server Error

When an unexpected server error occurs.

To capture a 500 error screenshot, trigger an internal server error in the application.

8. Tabi AI Assistant

Tabi () is an AI-powered chat assistant available on all authenticated pages. She appears as a floating owl button in the bottom-right corner.

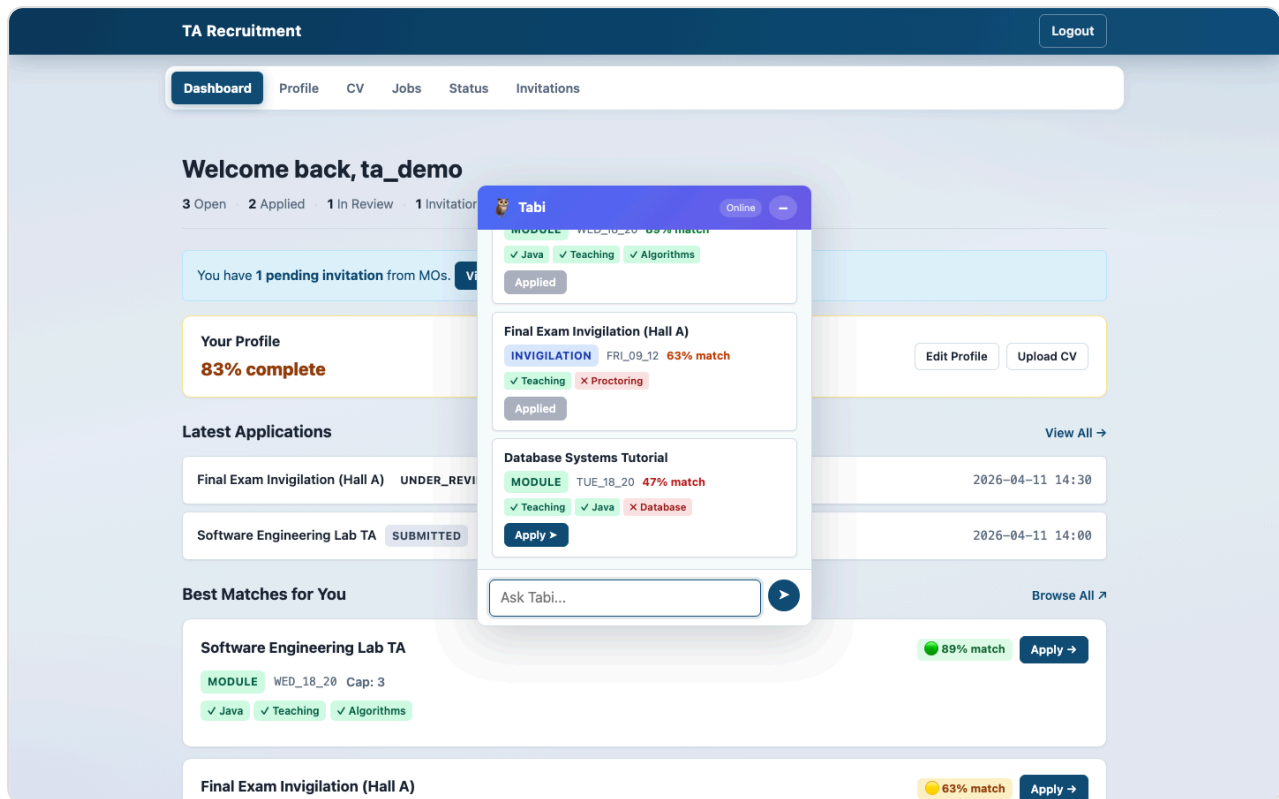
Features

- **Job recommendations:** Ask "What jobs match my skills?" to see open positions ranked by match score
- **Application status:** Ask "How are my applications doing?" to check statuses
- **Profile help:** Tabi detects missing profile fields and suggests what to fill in
- **MO applicant overview:** MOs can ask about applicant counts and status
- **Admin alerts:** Admins can ask about over-capacity jobs or unstaffed positions

How to Use

1. Click the floating 🦉 **button** on any page
2. The chat panel opens — type a question and press Enter
3. Tabi responds with text and interactive cards (job listings, status summaries, etc.)
4. Click links in cards to jump directly to relevant pages
5. Click the – **button** or anywhere outside to close

You can **drag** the button or the chat panel header to reposition them.



9. End-to-End Walkthrough

Follow these steps to verify the complete workflow:

Step-by-step

1. **Register as TA:** Go to `/register`, create a new TA account
2. **Complete profile:** Navigate to Profile, fill in all fields, add skills and availability
3. **Upload CV:** Go to CV, upload a `.txt` CV file
4. **Logout**, then login as `mo_demo`

5. **Post a job:** Click "+ Post New Job", fill the form, submit
6. **Logout**, then login as your TA account
7. **Browse jobs:** Go to Jobs, verify the new job appears with a match score
8. **Apply:** Click Apply on the job
9. **Check status:** Go to Status, verify application appears as SUBMITTED
10. **Logout**, login as `mo_demo`
11. **Review:** Go to Review Applications, find the TA's application
12. **Progress the application:** Under Review → Select
13. **Logout**, login as your TA account
14. **Verify:** Go to Status, verify status updated to SELECTED
15. **Logout**, login as `admin_demo`
16. **Check workload:** Verify the job appears in the workload table with correct counts

TA Recruitment

Logout

Dashboard
Profile
CV
Jobs
Status
Invitations

My application status

Refresh for the latest updates.

APPLICATION ID	JOB	MATCH	MISSING SKILLS	STATUS	FEEDBACK	LAST UPDATED	ACTION
A-DEMO-002	Final Exam Invigilation (Hall A)	50% Skills: 28/50 (Teaching matched; Proctoring missing) Schedule: 15/25 (not matched) Major: 15/15 Profile: 5/10	Proctoring	UNDER_REVIEW	Willing to complete proctoring training if selected.	2026-04-12 08:00	
A-DEMO-001	Software Engineering Lab TA	100% Skills: 44/50 (Java, Teaching, Algorithms matched) Schedule: 25/25 (matched) Major: 15/15 Profile: 5/10	None — all skills matched	SUBMITTED	Interested in SE lab; available WED evening.	2026-04-11 14:00	Withdraw